Topology-Correcting Differentiable Triangle-Soup Reconstruction via Persistent Homology

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July 9, 2026

Abstract

Resampling priors can steer *where* a differentiable triangle-soup reconstruction spends its triangle budget, but our controlled study of that channel (reported in Appendix A) found their topological value is carried largely by spatial *width*, and that no prior shape repairs one-dimensional features: concentrated priors manufacture phantom handles, spread priors never beat a random control. We move topology from the allocation channel into the *objective*: a differentiable persistence loss $L = \text{photometric} + \lambda(t)\, L_{\text{topo}}$, where L_{topo} is an optimally-matched distance between the live surface's persistence diagram and a fixed target diagram. Gradients flow through a pair-frozen backward pass: the matched birth/death simplices are re-expressed as closed-form circumradii of live sample positions — exact for the Gabriel simplices that constitute all significant pairs in practice — and a recruitment term restores the gradient that optimal matching provably lacks when a target feature is missing entirely. A single ratio knob ρ sets the loss scale by gradient-norm calibration; a curriculum proved unnecessary. On budget-starved synthetic scenes the loss is topology-specific where a norm-matched non-topological control is not: enclosed voids improve 4.0–7.9× and — the channel's distinguishing result — torus loops improve 2.3× with zero phantom handles across seeds, while the control collapses a loop. The two channels compose: loss plus the best resampling prior reaches 7.6× below baseline at the best Chamfer. Component counting (H0) is again matched by the generic control — a third channel agreeing that this class needs no topological information. All verdicts replicate, with no per-shape tuning, on three external genus-known meshes (2.1–10.4× at equal-or-better Chamfer). Prescription: correct topology in the loss, allocate with a wide prior, and combine.

1 Introduction

Differentiable rasterization reconstructs a triangle soup from images by following photometric gradients [1], periodically resampling triangles under a fixed budget. Photometric losses are topologically blind: reconstructions that differ in genus, connectivity, or enclosed-void structure can be indistinguishable to Chamfer and Hausdorff distances while the bottleneck distance between persistence diagrams separates them by an order of magnitude (Appendix A). We first attacked this through the *allocation* channel — biasing where the resampler spends triangles — in a controlled study reported with its key numbers in Appendix A. The verdict was mixed: a topological prior helps enclosed voids, but most of the gain is reproduced by any equally-wide non-topological prior, and

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no prior shape repairs one-dimensional features: concentrating manufactures phantom handles, and spreading never beats a random control on loops.

This paper moves topology into the *objective*. We add a differentiable persistence term to the training loss, so gradients act on the birth/death values of the reconstruction's topological features directly, rather than on where triangles respawn. Concretely:

- **A matched persistence loss with a pair-frozen backward** (§3): the forward pass is ordinary exact persistence (GUDHI alpha complex); the backward pass re-expresses each paired birth/death simplex as the closed-form circumradius of live surface samples. On our shapes, 100% of significant pairs are Gabriel-exact, so the frozen gradient is the true gradient almost everywhere.

- **A recruitment term for missing features**: optimal partial matching yields exactly zero gradient when a target feature has no live counterpart — the most important failure mode. Recruitment attaches the nearest live bar instead, restoring a descent direction; a controlled toy probes its known location-blindness, and an in-loop ablation shows the term is load-bearing — removing it collapses the loop-class win to baseline (§5).

- **One calibrated knob**: λ is set by a gradient-norm ratio ρ against the photometric loss. The response is flat over a 10× range of ρ, and a curriculum schedule adds nothing once calibrated — an ablation, not an assumption.

- **Channel-controlled evaluation**: every claim is tested against a norm-matched *non-topological* control acting through the identical channel (a repulsion on the same surface samples), inheriting the control discipline that made the allocation-channel study conclusive.

The loss is topology-specific for enclosed voids (4.0–7.9× error reduction at Chamfer parity) and, unlike every prior shape we tested, for loops (2.3×, zero phantom handles across seeds). The two channels compose (best-of-both 7.6×), and component structure (H0) is once again matched by the generic control — three channels now agree that this class carries no exploitable topological signal.

2 Related work

Differentiable rasterization and soup pipelines. OpenDR [2], neural mesh rendering [3], SoftRas [4] and nvdiffrast [5] established gradient flow from pixels to geometry; Gaussian splatting [6] and triangle splatting [7] trade connectivity for speed. The pipeline we build on is Diff-Soup [1], whose in-loop prune/respawn resampler is exactly the mechanism our allocation-channel study (Appendix A) steered with importance priors; here the resampler stays untouched and the objective changes instead. Vertex-domain optimizers such as large-steps preconditioning [8] are complementary — they change *how* vertices move, we change *why*.

Persistence in learning objectives. Persistent homology [9] with its stability guarantees [10] has been used as a trainable signal via differentiable sorting of filtration values [11], for topology-preserving image segmentation [12, 13], point-cloud and shape optimization [14, 15], and diagram-metric optimization with convergence analysis [16]; barcode-valued maps have since been given a general differential framework [17], and the per-step cost of persistence-guided descent attacked directly [18]. Our backward pass is in this family — gradients through the filtration values of paired simplices — with three pipeline-specific ingredients: an *alpha-complex* pairing over $\alpha \times$ area surface samples of a live triangle soup (rather than image grids or raw clouds), a Gabriel-exactness gate that certifies the circumradius re-expression, and the recruitment term for missing target features, which pure optimal matching cannot see.

Topology-aware geometry processing. Topology-aware reconstruction and repair predate differentiable pipelines [19–21], and budgeted remeshing has a long lineage [22–26]. Our contribution is orthogonal: the budget mechanics stay DiffSoup's; only the training signal becomes topology-aware.

3 Method: a matched persistence loss for triangle soups

3.1 Objective

DiffSoup trains with a photometric objective L_{photo} (opacity auxiliary + $0.8\,L_1 + 0.2\,\text{SSIM}$) [1]. We add

$$L \;=\; L_{\text{photo}} \;+\; \lambda(t)\,L_{\text{topo}}(V), \tag{1}$$

where V are the soup's vertex positions and L_{topo} compares the persistence diagram of the live surface to a fixed *target bundle* $\mathcal{T}$: the significant finite bars of the ground-truth shape's diagram, computed once at the same sampling density M the loss will see (alpha birth/death values shift with point spacing, so density matching is a correctness requirement, not a convenience) and normalized by the target's bounding-box diagonal, which is locked for all live evaluations.

3.2 Live diagram and pair-frozen backward

At each refresh we draw $M{=}2048$ surface samples $X_j = \sum_i w_{ij}\,V[F(f_j), i]$ with faces f_j drawn $\sigma(\alpha)\times$area-weighted and barycentric weights w_{ij} frozen; X is recomputed from live V every step, so gradients reach vertices while the combinatorics stay fixed. The alpha complex of X yields persistence pairs (birth simplex, death simplex) per dimension. For a Gabriel simplex the alpha filtration value is its squared circumradius, so each paired bar's coordinates are closed-form functions of a few sample positions — half edge length (H0 deaths), triangle circumradius (H1 deaths, H2 births), tetrahedron circumradius (H2 deaths) — all exactly differentiable. We verify, per pairing, that the re-expressed value matches the filtration value (relative 10^{-6}); non-Gabriel simplices would contribute value without gradient, but on all six benchmark shapes 100% of significant pairs are Gabriel-exact. The pairing is treated as locally constant — the standard pair-frozen device [11, 16] — and refreshed every $K{=}10$ steps and, necessarily, immediately after every resampling event, which replaces the vertex and face tensors wholesale.

3.3 Matching, diagonal pressure, and recruitment

Let D be the live significant bars and $\mathcal{T}$ the target bars of one dimension. An optimal partial matching (Hungarian on the standard augmented cost; equivalently W_2) splits bars into matched pairs, unmatched live, and unmatched target:

$$L_{\text{topo}} = \sum_{(i,j)\,\text{matched}} \left[(b_i - b_j^*)^2 + (d_i - d_j^*)^2\right] \;+\; \sum_{i\,\text{unmatched}} \left(\tfrac{d_i - b_i}{2}\right)^2 \;+\; L_{\text{recruit}}. \tag{2}$$

The first term pulls matched features to their target coordinates; the second pushes spurious features to the diagonal. The third exists because optimal matching has a blind spot: when a target feature is entirely missing, the cheapest transport diagonalizes both sides and the gradient is exactly zero — precisely in the most important failure case. Recruitment attaches each unmatched target bar to the nearest not-yet-matched live bar (sub-threshold bars allowed) and pulls it toward the target. This deliberately departs from a metric-faithful W_2; §5 probes its known pathology (location-blindness) directly.

3.4 Calibration instead of curriculum

$\lambda(t)$ ramps linearly from 20% to 50% of training, with peak set once by a gradient-norm ratio at the first active step: $\lambda_{\text{peak}} = \rho\,\|\nabla_V L_{\text{photo}}\|/\|\nabla_V L_{\text{topo}}\|$. The single knob ρ is dimensionless; ρ=0.1 throughout. The tail error is flat across $\rho \in \{0.03, 0.1, 0.3\}$, and an uncalibrated-schedule ablation (constant λ from step one) matches the ramped version — calibration, not scheduling, is what makes the knob safe.

3.5 Conditions

C0: baseline (photometric only). C1: $+L_{\text{topo}}$. C2: + a short-range repulsion on the *same* frozen samples, calibrated with the *same* ρ — a norm-matched non-topological control through the identical channel (C2g: a gentler radius). C3: C1 without the ramp. C5: C1 combined with the best prior of our allocation-channel study (the spread topological resampling field B4; defined in Appendix A). C6: C1 with the recruitment term removed (pure matching + diagonal, all else identical) — does recruitment matter in-loop, or only for cold-start repair? The hook adds one loss line to the training loop; with the flag off, no new code executes, and a ρ=0 run is divergence-equivalent to baseline re-runs under CUDA nondeterminism.

4 Experimental setup

Scenes and budgets. Synthetic COLMAP scenes with analytically known topology, rendered from 24–72 Fibonacci-sphere viewpoints at 200–256 px (the loader's standard every-8th test holdout leaves 21–63 training views): sphere and cube (void class H2, β=$(1,0,1)$, budget N=1200), torus (loop class H1, $(1,2,1)$, N=700), two spheres (component class H0, $(2,0,2)$, N=700), and genus-2 double torus as a stretch case ($(1,4,1)$, N=2000). Budgets are the allocation study's headroom points (Appendix A) — tight enough that the baseline's topology is imperfect, so there is something to win. Three external, genus-known, watertight meshes probe generality beyond the analytic family (§5.3): *spot* (genus 0) and *bob* (genus 1) from the Crane model repository (CC0), and the *fandisk* CAD benchmark (genus 0) — rendered with the same pipeline (48 views, 200 px), class budgets carried over unchanged (N=1200 voids, N=700 loops), and a seed-0 C0 pilot confirming baseline imperfection (tail bottleneck .043–.054, i.e. inside the analytic shapes' band).

Protocol. 2,500 steps; resampling every 100 steps (untouched); ρ=0.1, ramp 0.2:0.5, refresh K=10, M=2048; decisive shapes (sphere, torus) run 5 seeds with the full condition set (C0/C1/C2/C3/C5, plus the recruitment ablation C6), replication shapes (cube, two spheres) 3 seeds (C0/C1/C2), double torus 2 seeds. The torus restricts the loss to H1: its own void bar is sub-threshold at M=2048, and a loss must not act on features it cannot reliably measure at its sampling density. For the same reason the double torus targets its two measurable loops (of four) — matching what the reconstruction itself can express at any feasible budget. The generality trio runs C0/C1/C2 with three seeds; bob, unlike the thin torus, has both its loop pair *and* its void significant at M=2048 (fat tube), so it runs the full unrestricted bundle.

Table 1: Tail bottleneck to target (lower is better); $\times$ = reduction vs C0. Verdict requires C1 < C0 and C1 < every control at Chamfer parity.

shape (dim)	C0	C1 loss	C2 control	C3 no-ramp	C5 loss+prior
sphere (H2)	.0623±.0063	**.0156±.0013** (4.0×)	.1372 (worse)	.0151±.0004	**.0082±.0007** (7.6×)
cube (H2)	.0582±.0046	**.0074±.0011** (7.9×)	.1346 (β_2 1→0)	—	—
torus (H1)	.0424±.0031	**.0180±.0016** (2.3×)	.0457 (β_1 2→1)	.0155±.0015	**.0133±.0004** (3.2×)
two_spheres (H0)	.0065±.0008	.0007±.0002 (9.5×)	.0008 (ties C1)	—	—
double_torus (H1)	.0264±.0000	.0264±.0000 (saturated)	.0264±.0000	—	—

Metrics and verdict. Primary: bottleneck distance to the target diagram in the shape's discriminating dimension, tail-averaged over the last trajectory dumps, seed-averaged (20k evaluation samples, target-normalized). Secondary: significant-feature counts (phantom check) and Chamfer parity. *Verdict rule:* C1 must beat C0 **and** every norm-matched control at Chamfer parity — a topological win a dumb regularizer can match is not a topological win. CUDA training is not bit-reproducible (baseline re-runs diverge at the first resample), so no bit-level claims are made anywhere.

5 Results

5.1 Gate: the loss alone repairs topology

Before integration, the loss (no photometric term, Adam directly on point positions) must repair defective clouds. It does, on all four probes (Fig. 1): a punctured sphere's void birth is pulled to target (8.8× bottleneck reduction; the hole closes), a spurious handle is pushed to the diagonal and dies (H1 bar eliminated; β restored), mis-spaced components are pulled to the target separation (1229×), and — the recruitment probe — a bridged merge with *no* significant live H0 bar is severed (450×, β_0 1→2). The predicted location-blind pathology did not materialize, for a mechanism reason worth stating: tearing a 2-manifold shell cannot grow a component bar past the local detour scale (the tug is abandoned), while cutting the 1-D bridge raises the recruited bar's death monotonically toward target — the runaway concentrates on the only profitable direction.

5.2 The C-matrix

Table 1 reports tail bottleneck (mean±sd over seeds) in each shape's discriminating dimension; Chamfer in the text where it matters; Fig. 2 shows the full training trajectories for the two decisive shapes.

Voids (H2): topology-specific, large. Sphere 4.0× (16.2σ Welch vs C0), cube 7.9× (18.5σ), both at equal-or-better Chamfer (sphere 0.46 vs 0.50; cube 0.92 vs 1.05). The norm-matched control is not merely weaker — it is *destructive*: it pins the void's death at its repulsion equilibrium (bottleneck 2.2× worse than baseline, identical across seeds) and on the cube erases the void outright. Nor is this an artifact of control strength: a gentle variant at half the push radius still lands 1.6× *worse* than baseline (.0972 vs .0623) — the sphere ordering is C1 < C0 < C2g < C2, so no tested repulsion strength reproduces any of the gain (Fig. 3).

Loops (H1): the channel's distinguishing result. In the allocation-channel study (Appendix A) no prior shape won this class — concentration manufactured phantom handles; spreading never beat a random control. The loss cuts torus H1 error 2.3× with *zero phantom handles in all five seeds* (significant-count trajectories flat at the true β_1=2, Fig. 4), while the control collapses one loop. Metric pressure on birth/death values suits 1-D features in a way budget allocation does not.

Components (H0): not topological — third channel, same answer. C1 reduces the (already small) baseline error 9.5×, but the generic control matches it on the diagram metric (.0008 vs .0007); the topological increment appears only in Chamfer (0.54 vs 1.94). Together with the allocation-channel results this is now a statement about the feature class: component structure is helped by any spreading pressure, not by topological information.

Composition (C5). Loss + spread prior is best everywhere tested — sphere .0082 (7.6× vs C0, ≈2× vs C1 alone, ≈4.6× below the prior alone), torus .0133 — with the best sphere Chamfer as well; on the torus its Chamfer (.502) trails only the no-ramp arm (.495). Allocation and value-tuning are complementary mechanisms, not substitutes.

Stretch case (genus 2): saturated, honestly. On the double torus every arm — baseline, loss, control — returns a bottleneck of exactly .0264 (sd 0 over seeds), and Wasserstein is equally pinned (.0527–.0528): the two tube loops are unexpressable at feasible budgets (replicating the allocation study's floor, Appendix A), and they dominate both diagram metrics regardless of guidance. The loss adds no phantoms there (β_1-significant count = 2 in every run) and its Chamfer is marginally best; the case measures the *representation's* ceiling, not the channel's.

Ablations. ρ-response flat over $\{0.03, 0.1, 0.3\}$ (.0181/.0172/.0177 on sphere); C3 (no curriculum) ≈ C1 on both decisive shapes. Overhead: ≈45–55 s per 2,500-step run (refresh 174 ms at M=2048, every 10 steps).

Recruitment is load-bearing for loops (C6). Removing the recruitment term (pure matching + diagonal, all else identical) collapses the torus win entirely: .0411 ± .0038, statistically baseline (0.6σ from C0, 12.4σ worse than C1) at baseline Chamfer — though the loops still *exist* (#sig H1 = 2 in every seed; Fig. 2, right: the C6 curve rides the baseline). The run logs locate the mechanism precisely: at the loss's own sampling density, one torus loop is significant at every refresh while the other sits chronically *below* the significance threshold — in all five seeds, at all 200 refreshes — so matching and diagonal pressure act only on the loop the live diagram already measures, and recruitment is the sole gradient path to the other. It is not a cold-start device; on this class it *is* the mechanism. On the sphere the void is always live and matched, recruitment never fires (0 of 200 refreshes, all seeds), and C6 is loss-identical to C1 by construction — its tail (.0166 vs C1's .0156, 1.5σ) doubles as a measurement of pure run-to-run (CUDA) noise.

5.3 Generality: external genus-known meshes

Everything above is on one analytic scene family. We therefore repeat the verdict protocol — C0/C1/C2, three seeds, class budgets, ρ, ramp and sampling unchanged — on three external watertight meshes of known genus (§4): *spot* (genus 0, organic), *bob* (genus 1, organic), and the *fandisk* CAD benchmark (genus 0). No per-shape tuning of any kind: the headroom budgets transfer from the feature class, and bob runs the full unrestricted bundle. All three **pass** the verdict rule at equal-or-*better* Chamfer (Table 2, Fig. 5), and the control's failure modes replicate

Table 2: Generality wave: tail bottleneck to target (mean±sd, 3 seeds); × = reduction vs C0; σ = Welch. Same verdict rule as Table 1.

shape (dim)	C0	C1 loss	C2 control	verdict
spot (H2)	.0521±.0081	**.0237±.0000** (2.2×)	.0652 (β_2 1→0)	PASS (6.1σ)
bob (H1)	.0426±.0012	**.0208±.0008** (2.1×)	.0437 (β_1 2→1)	PASS (26.1σ)
fandisk (H2)	.0538±.0022	**.0052±.0008** (10.4×)	.0571 (β_2 1→0)	PASS (35.7σ)

exactly: repulsion erases the void on both H2 shapes (β_2 1→0, all seeds) and collapses a loop on the genus-1 shape (β_1 2→1, all seeds).

Three observations. *The H1 claim survives the family change*: bob halves loop error with zero phantom handles in every seed (26.1σ), replicating torus. *Sharp-featured CAD benefits most*: fandisk's 10.4× exceeds every analytic-family effect. *Complex organic geometry exposes a count-level limit*: spot's *baseline* manufactures a phantom void in two of three seeds (final #sig H2 = $2, 1, 2$); the loss clears it in two seeds but retains one spurious bar in the third $(1, 2, 1)$ — value repair is reliable, count repair on sub-budget organic detail (legs, horns at N=1200) is an improvement, not a guarantee. Spot's C1 tail is also pinned across seeds (sd 0 at .0237), suggesting a geometric floor of the same kind as the double torus, far milder in degree.

6 Mechanism and discussion

Sparse tugs vs. budget allocation. Each significant pair back-propagates through at most a handful of sampled points, so the loss delivers *precise but sparse* pulls, iterated by re-pairing — the toy probes show surgery "walking" around a defect a few points at a time. The resampling prior is the opposite: broad, indirect, budget-limited. The results sort cleanly along this axis: 1-D loops, which the prior could only over- or under-tessellate, respond to precise value pressure (H1 win); wide 2-D shells benefit from both (H2: prior helps, loss helps more, composition best); 0-D component structure needs neither — any spreading pressure suffices (H0 null, three channels).

Why the control damages topology. The norm-matched repulsion imposes a length-scale floor on the sampled surface; the void's death time pins at that equilibrium (identical across seeds) and thin structures collapse (β_2 loss on the cube, β_1 loss on the torus). This sharpens the control argument: spending the same gradient budget through the same channel *without* topological information is not neutral — it is harmful — so C1's wins cannot be an artifact of extra regularization.

Measurability at the loss's density. A persistence loss can only enforce features that clear the significance floor at its sampling density M: the torus void and the double torus's two tube loops do not, at M=2048. We treat this as a principle rather than a nuisance: the target bundle is built (and audited) at the same M, features below the floor are excluded from the objective, and the double-torus stretch case is evaluated against its two measurable loops — which is also all the reconstruction itself can express at feasible budgets. Raising M buys resolution at linear-ish persistence cost; the trade is explicit and logged at bundle-build time.

Recruitment is landscape-guided in practice — and it is the mechanism for loops. Its location blindness — pull the nearest bar anywhere — did not misfire on our geometry because unproductive tears self-extinguish (their bar's death saturates at the local detour scale) while productive cuts run away toward the target. The caveat stands for target features larger than any achievable detour; the probe and its mechanism are reported so the term is used with eyes open. The C6 ablation sharpens its role beyond cold-start repair: a loop that never clears the live significance floor at M receives metric pressure *only* through recruitment, at every refresh — matching alone reduces the torus to baseline. Recruitment is how the loss reaches below its own significance floor on the live side.

Prescription. Correct topology in the *loss*; allocate budget with a *wide* prior; combine for the best of both. Curriculum schedules are unnecessary once the loss scale is calibrated against the photometric gradient.

7 Limitations

Synthetic, closed, single-machine. All quantitative claims are on synthetic renders of closed surfaces with known topology, on one GPU. The scene family is analytic plus three external genus-known meshes (§5.3) — still short of a public shape-set benchmark — and a real (open-surface, clinical) demonstration remains future work; open surfaces add a genuine research obstacle — scan boundaries contaminate H1, and no clean ground-truth diagram exists.

Density floor. The loss cannot see features below the significance threshold at its sampling density (torus void; two of four double-torus loops at M=2048). The bundle audit makes this visible, not invisible — but it is a real ceiling on what the objective can enforce.

Fixed target diagram. The target bundle presumes known ground-truth topology (true in reconstruction-from-scans settings where a prior scan or template exists; not in blind capture). Template-free variants (e.g., losses on diagram *statistics*) are untested here.

Control strength. The repulsion control is aggressive (it destroys topology at the calibrated magnitude); a gentler variant at half the radius is still 1.6× worse than baseline (sphere: .0972 vs .0623), so the verdicts do not hinge on control tuning — but control *design* for loss-channel studies deserves more variety than one family at two strengths.

Nondeterminism. DiffSoup's CUDA training is not bit-reproducible — even baseline re-runs diverge in resampling decisions — so all statements are seed-averaged; per-run reproductions will differ in detail.

Deferred arms. Curvature-weighted spatial masking of the loss (C4) and the dental showcase are designed but not run; both are gated on a small weight-mask extension to the loss state.

A The allocation-channel study, in brief

This appendix reports, with their numbers, the three findings of our allocation-channel study — controlled resampling-prior experiments on the same pipeline, scenes, and budgets, conducted before moving topology into the objective — that this paper builds on.

(1) Geometric metrics are topology-blind. Three constructed cases each pit a topologically correct candidate A against a wrong one B whose geometry is bisection-matched, making Chamfer *equal by construction* (40k samples, metrics normalized by the ground-truth bounding diagonal; Table 3). The discriminating-dimension bottleneck separates every pair — by 35–40× in the H0/H2 cases — while in the first two cases Hausdorff$_{95}$ actually *prefers* the topologically wrong candidate. Chamfer cannot tell a benign bump from a destroyed void; the persistence metric can.

Table 3: Blindness probes: A topologically correct, B wrong, geometry bisection-matched so Chamfer is equal by construction. Bottleneck to ground truth in the discriminating dimension.

case (disc. dim)	Chamfer% A = B	bott. A	bott. B	ratio
thin handle, genus 0→1 (H1)	0.916	≈0	.0302	—
bridge merges two components (H0)	0.628	.0014	.0574	40×
punctured enclosed void (H2)	0.977	.0040	.1385	35×

(2) The prior arms, and why voids are "width-primary". The study biases DiffSoup's own resampler with a precomputed importance field: persistence-localized Gaussian splats at each target feature's death-simplex location ($\sigma_f = \sqrt{\text{death}_f}$), applied at budget parity as prune protection plus respawn bias. Arms: B0 baseline; B2 the concentrated field; B4 the same mass *spread* over the feature's support ($\sigma \to s\sigma$, $w \to \text{life}/s^2$, mass-preserving, s=3); B1 = B2 applied at initialization only; B3/B5 random-center non-topological controls width-matched to B2/B4 respectively. C5 in this paper reuses exactly the B4 field. On the void class (Table 4; N=1200, five paired seeds) B4 cuts the bottleneck 33–53% below baseline — but the width-matched random control B5 recovers most of that gain, leaving only a small, shape-dependent topological residual (cylinder 4.6σ, sphere 2.2σ, cube tie); one-shot placement washes out entirely (B1 ≈ B0, .0546 vs .0535 on the sphere). Hence the verdict quoted in §1: the prior's topological value is carried largely by its spatial *width*.

Table 4: Companion study, void class (H2), N=1200, five paired seeds: tail bottleneck to target (lower is better). Spread beats concentrated; the width-matched random control (B5) recovers most of the spread gain.

shape	B0	B2 concentrated	B3 narrow rand.	B4 spread	B5 wide rand.
sphere	.0535	.0438	.0440	**.0357**	.0374
cube	.0579	.0409	.0403	.0273	**.0270**
cylinder	.0545	.0440	.0477	**.0360**	.0402

(3) No prior shape repairs loops. On the torus (N=700, five paired seeds, true β_1=2) every arm fails the topology-specificity test: the best condition is the *non-topological* wide control B5 (.0267), ahead of spread-topological B4 (.0316), baseline (.0409), and narrow random B3 (.0397); the concentrated topological field B2 is worst (.0425) *and* manufactures phantom handles — final significant-H1 count 4.4 against the true 2.0, with the worst Chamfer (1.32 vs baseline 1.01). Spreading merely stops the harm; it never beats the random control. That failure is the opening premise of this paper, and the class the loss channel wins (§5.2, Table 1).

References

[1] Kenji Tojo, Bernd Bickel, and Nobuyuki Umetani. DiffSoup: Direct differentiable rasterization of triangle soup for extreme radiance field simplification. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. arXiv:2603.27151.

[2] Matthew M. Loper and Michael J. Black. OpenDR: An approximate differentiable renderer. In *European Conference on Computer Vision (ECCV)*, pages 154–169, 2014.

[3] Hiroharu Kato, Yoshitaka Ushiku, and Tatsuya Harada. Neural 3D mesh renderer. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3907–3916, 2018.

[4] Shichen Liu, Tianye Li, Weikai Chen, and Hao Li. Soft rasterizer: A differentiable renderer for image-based 3D reasoning. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 7708–7717, 2019.

[5] Samuli Laine, Janne Hellsten, Tero Karras, Yeongho Seol, Jaakko Lehtinen, and Timo Aila. Modular primitives for high-performance differentiable rendering. *ACM Transactions on Graphics*, 39(6):194:1–194:14, 2020.

[6] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3D Gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 42(4):139:1–139:14, 2023.

[7] Jan Held, Renaud Vandeghen, Adrien Deliège, Abdullah Hamdi, Silvio Giancola, Anthony Cioppa, Andrea Vedaldi, Bernard Ghanem, Andrea Tagliasacchi, and Marc Van Droogenbroeck. Triangle splatting for real-time radiance field rendering. arXiv:2505.19175, 2025.

[8] Baptiste Nicolet, Alec Jacobson, and Wenzel Jakob. Large steps in inverse rendering of geometry. *ACM Transactions on Graphics*, 40(6), 2021.

[9] Herbert Edelsbrunner, David Letscher, and Afra Zomorodian. Topological persistence and simplification. *Discrete & Computational Geometry*, 28(4):511–533, 2002.

[10] David Cohen-Steiner, Herbert Edelsbrunner, and John Harer. Stability of persistence diagrams. *Discrete & Computational Geometry*, 37(1):103–120, 2007.

[11] Rickard Brüel Gabrielsson, Bradley J. Nelson, Anjan Dwaraknath, and Primoz Skraba. A topology layer for machine learning. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 108 of *Proceedings of Machine Learning Research*, pages 1553–1563, 2020.

[12] Xiaoling Hu, Fuxin Li, Dimitris Samaras, and Chao Chen. Topology-preserving deep image segmentation. In *Advances in Neural Information Processing Systems 32 (NeurIPS)*, 2019.

[13] James R. Clough, Nicholas Byrne, Ilkay Oksuz, Veronika A. Zimmer, Julia A. Schnabel, and Andrew P. King. A topological loss function for deep-learning based image segmentation using persistent homology. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44 (12):8766–8778, 2022.

[14] Rickard Brüel-Gabrielsson, Vignesh Ganapathi-Subramanian, Primoz Skraba, and Leonidas J. Guibas. Topology-aware surface reconstruction for point clouds. *Computer Graphics Forum*, 39(5):197–207, 2020.

[15] Adrien Poulenard, Primoz Skraba, and Maks Ovsjanikov. Topological function optimization for continuous shape matching. *Computer Graphics Forum*, 37(5):13–25, 2018.

[16] Mathieu Carrière, Frédéric Chazal, Marc Glisse, Yuichi Ike, Hariprasad Kannan, and Yuhei Umeda. Optimizing persistent homology based functions. In *International Conference on Machine Learning (ICML)*, volume 139 of *Proceedings of Machine Learning Research*, 2021.

[17] Jacob Leygonie, Steve Oudot, and Ulrike Tillmann. A framework for differential calculus on persistence barcodes. *Foundations of Computational Mathematics*, 22:1069–1131, 2022.

[18] Arnur Nigmetov and Dmitriy Morozov. Topological optimization with big steps. *Discrete & Computational Geometry*, 2024. doi: 10.1007/s00454-023-00613-x.

[19] Andrei Sharf, Thomas Lewiner, Gil Shklarski, Sivan Toledo, and Daniel Cohen-Or. Interactive topology-aware surface reconstruction. *ACM Transactions on Graphics*, 26(3):43:1–43:10, 2007.

[20] Tao Ju. Robust repair of polygonal models. *ACM Transactions on Graphics*, 23(3):888–895, 2004.

[21] Marco Attene. A lightweight approach to repairing digitized polygon meshes. *The Visual Computer*, 26(11):1393–1406, 2010.

[22] Hugues Hoppe. Progressive meshes. In *Proceedings of SIGGRAPH 96*, pages 99–108, 1996.

[23] Michael Garland and Paul S. Heckbert. Surface simplification using quadric error metrics. In *Proceedings of SIGGRAPH 97*, pages 209–216, 1997.

[24] Pierre Alliez, David Cohen-Steiner, Olivier Devillers, Bruno Lévy, and Mathieu Desbrun. Anisotropic polygonal remeshing. *ACM Transactions on Graphics*, 22(3):485–493, 2003.

[25] Mario Botsch and Leif Kobbelt. A remeshing approach to multiresolution modeling. In *Eurographics/ACM SIGGRAPH Symposium on Geometry Processing (SGP)*, 2004.

[26] Marion Dunyach, David Vanderhaeghe, Loïc Barthe, and Mario Botsch. Adaptive remeshing for real-time mesh deformation. In *Eurographics 2013 — Short Papers*, 2013.

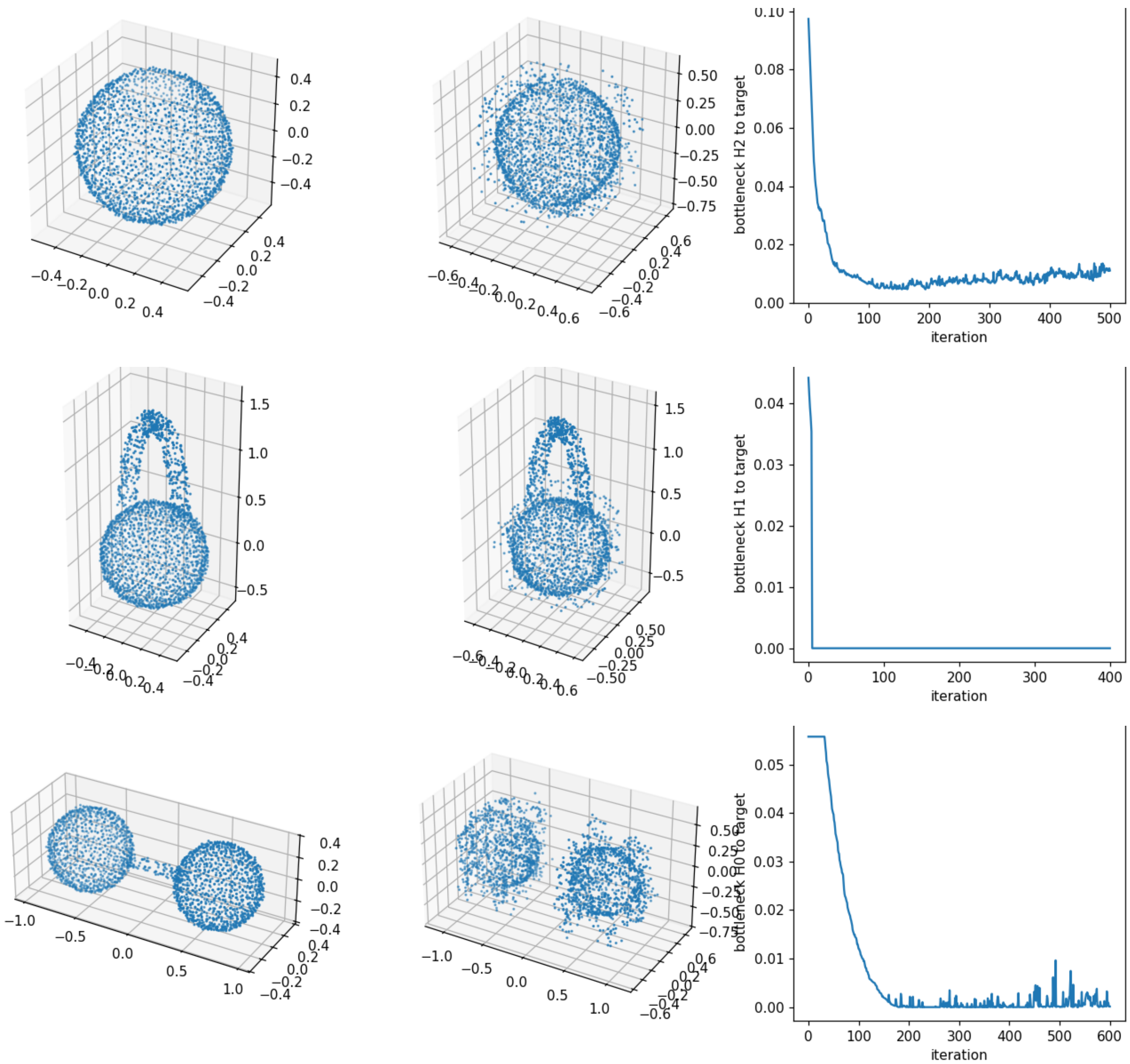

Figure 1: The stage-3a gate: the loss alone (Adam on raw point positions, no photometric term) repairs defective clouds. Each row shows the initial cloud, the final cloud, and the bottleneck-to-target trajectory in the discriminating dimension — one row per mechanism of the matched loss (§3). *Top* (T1): punctured sphere; the matched birth term pulls the void closed (8.8× reduction). *Middle* (T2): spurious handle; the diagonal term kills the extra H1 bar outright. *Bottom* (T4): bridged merge with no significant live H0 bar; recruitment severs the bridge (450×, β_0 1→2) — the case where plain optimal matching provably has zero gradient. The mis-spaced components probe (T3, 1229×) is omitted for space.

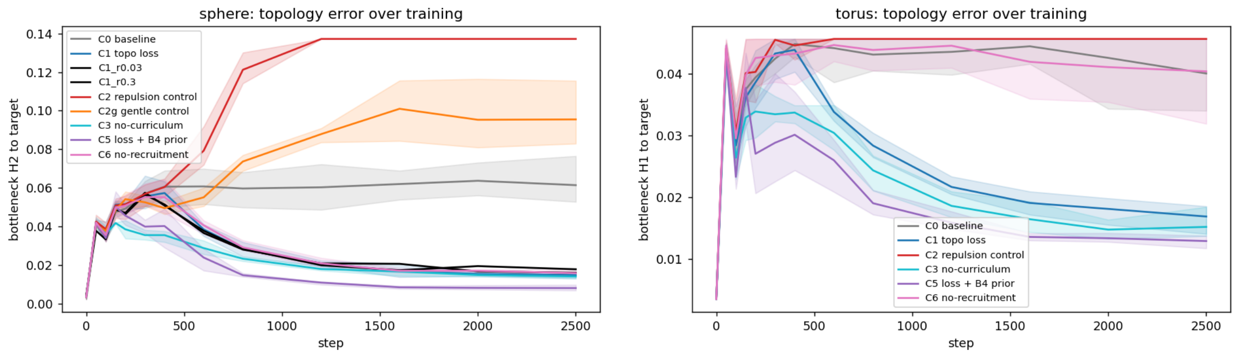

Figure 2: Topology error over training: bottleneck to target in the discriminating dimension (line = seed mean, band = seed min–max). *Left*, sphere (H2): every loss arm — C1, C3, C5, C6, and both ρ ablation variants — converges to a fraction of baseline, while the norm-matched repulsion control C2 is *destructive* and even its gentle variant C2g ends worse than baseline. *Right*, torus (H1), the class no resampling prior won: same ordering, with C5 (loss + spread prior) best — except the no-recruitment ablation C6, which rides the baseline (see *Ablations*).

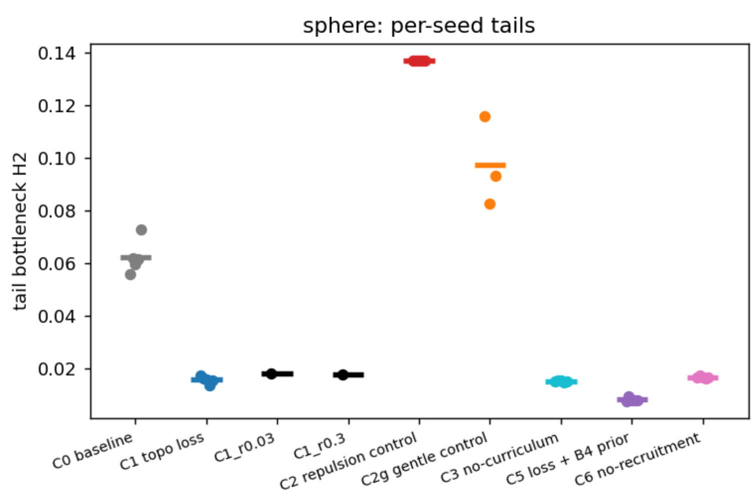

Figure 3: Sphere, per-seed tail bottlenecks (dots) with arm means (bars). Every loss arm sits far below baseline with tight seed spread; the two control strengths bracket baseline from *above*, so no tested repulsion strength reproduces any of the gain; the ρ ablation arms are indistinguishable from ρ=0.1.

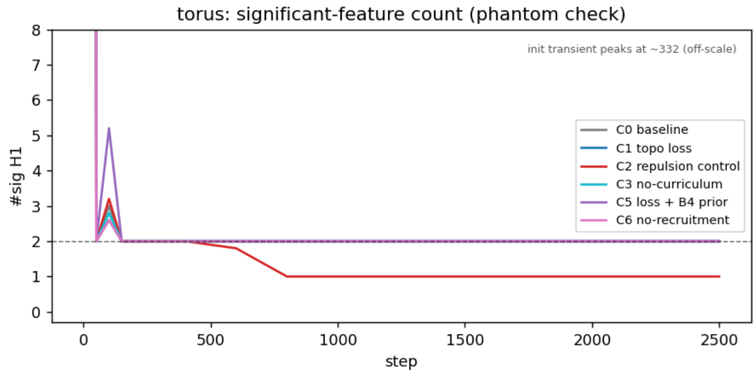

Figure 4: Torus phantom check: significant-H1 count over training (seed means; dashed line: true β_1=2). After a brief resampling transient near step 100, every loss arm sits exactly on two significant loops for the rest of training — zero phantom handles — while the repulsion control (C2) collapses a loop by step ∼800 and never recovers it. All arms share the off-scale random-init transient (annotated).

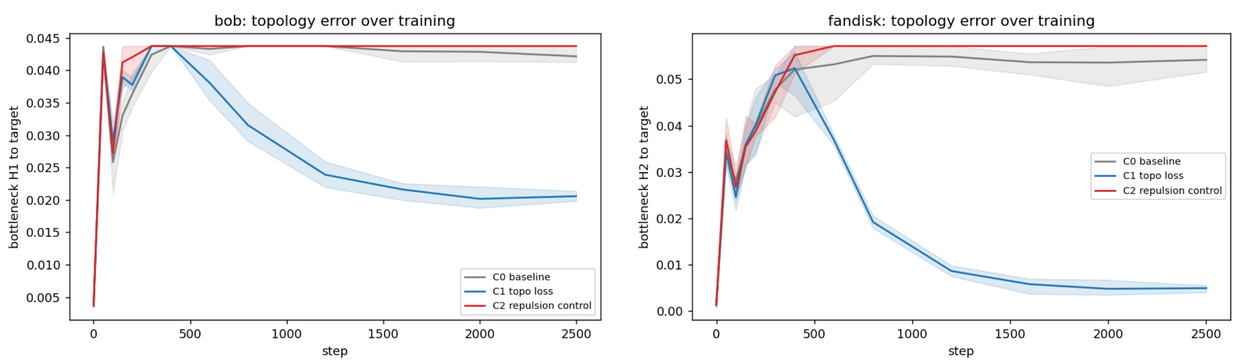

Figure 5: Generality trajectories (line = seed mean, band = seed min–max). *Left*: bob (genus 1) — the H1 result off the analytic family: the loss halves the loop error with #sig H1 = 2 in every seed, while the control rides the baseline after collapsing a loop. *Right*: fandisk (CAD) — the largest effect measured in this study (10.4×).